

Online Retail Sales Performance Analysis

Evaluating overall revenue trends, product performance, geographic distribution, and seasonal purchasing patterns to guide operational strategy.

DATASET SCOPE

1.0M+

Transactions Analyzed

OBSERVATION PERIOD

2009 – 2011

UCI Machine Learning Repository

CORE BUSINESS GOAL

Transform historical transaction data into strategic decisions that improve revenue capture, inventory allocation, and geographic expansion.

 Executive Presentation

Dataset Profile & Processing Methodology



Online Retail II

UCI Repository Dataset

TOTAL RECORDS

1,000,000+

TIME PERIOD

Dec 2009 – Dec 2011

TRACKED ATTRIBUTES

Invoice • Stock Code • Description •
Quantity • Invoice Date • Price • Customer
ID • Country

Analytical Pipeline & Data Hygiene

- 01

Deduplication & Cleaning
Removed duplicate order rows and missing descriptions.
- 02

Returns & Cancellation Isolation
Filtered negative quantities and cancelled invoice codes.
- 03

Revenue Formulation
Computed net revenue as $\text{Quantity} \times \text{Unit Price}$ per transaction line.
- 04

Outlier & Wholesale Screening
Isolated extreme bulk purchases to prevent metric skew.
- 05

Multi-Dimensional Analysis
Segmented results across time, geography, and SKU categories.

Key Findings: What Management Should Know

DOMINANT FACTOR

 UK Market Dominance

The UK generates over 80% of total online sales revenue, highlighting massive reliance on a single domestic market.

Strategic Implication:
High exposure to domestic market fluctuations requires expansion planning.

 November Sales Surge

November yields the highest monthly revenue consistently across both observed years due to holiday prep.

Peak Demand Driver

 High-Impact Top SKUs

Top products like *Regency Cakestand 3 Tier* contribute disproportionately to total revenue.

High SKU Concentration

 Geographic Clustering

International revenue is highly concentrated within a small cluster of Western European countries.

Regional Clustering

 Bulk Order Volatility

Occasional ultra-large wholesale orders significantly shift overall monthly volume and average order values.

Wholesale Outlier Impact

November Sales Peak & Seasonal Trajectory

PEAK MONTHLY REVENUE

Annual High

£1.50M

November 2011 Peak Sales (Follows £1.46M in Nov 2010)

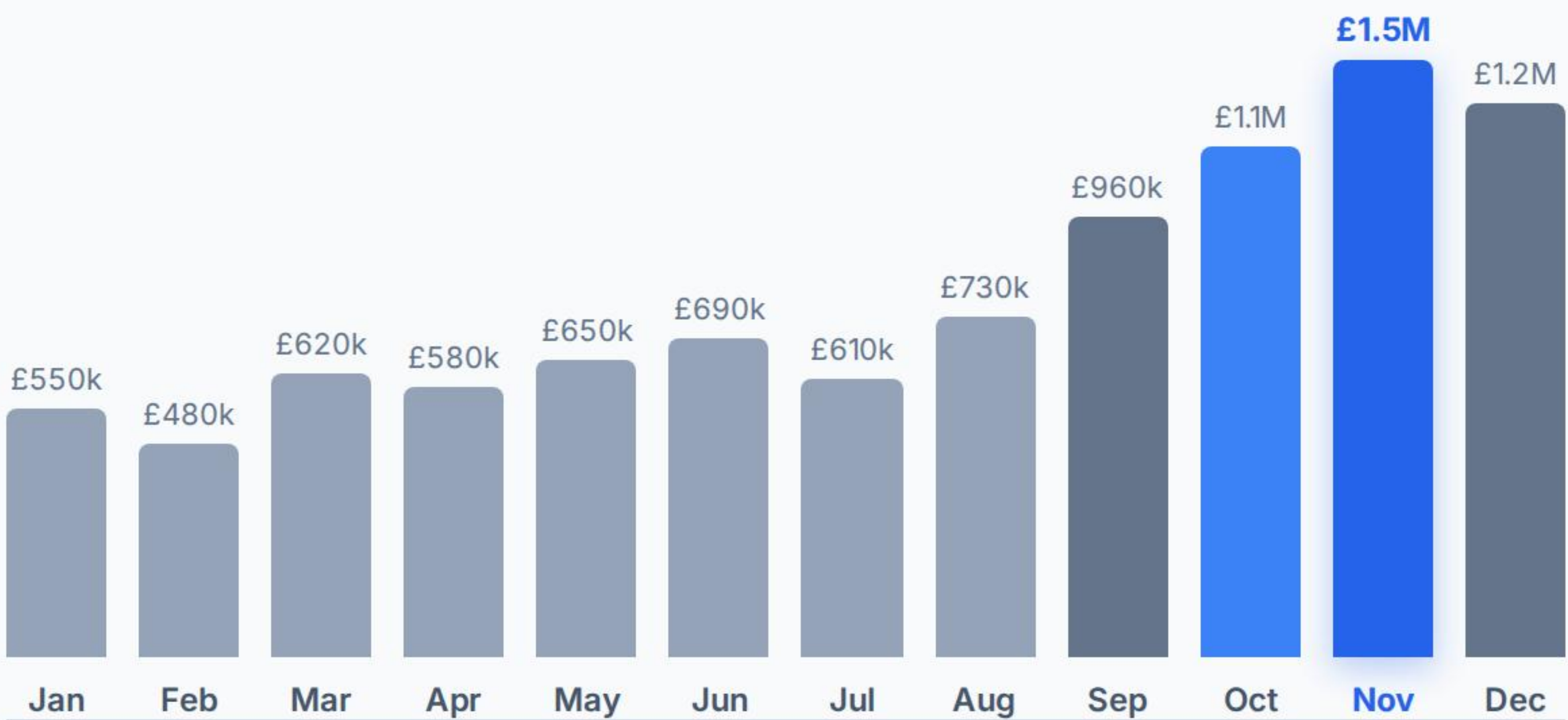
- ✓ **Predictable Annual Spike**
Revenue surges sharply in Q4 as retailers and consumers stock up ahead of the holiday season.
- ✓ **Inventory & Operational Bottlenecks**
Fulfillment capacity and supplier lead times must be locked in by August/September to avoid stock-outs.

Key Takeaway: November accounts for over 2x average monthly volume. Strategic inventory staging is critical.

Monthly Sales Distribution (£)

Illustrative seasonal curve across 12-month cycle

■ Nov Surge



3 Actionable Recommendations

HIGH PRIORITY

01

Prepare for Seasonal Peak Demand

Ramp up inventory buffers and lock in logistics fulfillment capacity prior to Q4 demand acceleration.

Owner: Sales & Operations
KPI: Nov Revenue & Stock-out Rate

HIGH PRIORITY

02

Mitigate Geographic Concentration

Protect core UK revenue while scaling targeted digital marketing into top European international markets.

Owner: Sales & Marketing
KPI: International Growth %

HIGH PRIORITY

03

Prioritize High-Performing SKUs

Maintain stock availability and dedicated promotional campaigns for top 10% revenue-generating items.

Owner: Product & Merchandising
KPI: Product Revenue & Availability

Expected Business Impact Matrix

Strategic Action	Expected Impact	Success Measure
Prepare for Seasonal Demand	Capture peak-period sales volume and minimize stock-out losses during Q4	November Revenue & Stock-out Rate
Expand Selected European Markets	Reduce geographic revenue concentration and UK market vulnerability	International Revenue Growth
Prioritize High-Performing SKUs	Improve product availability and maximize total margin opportunities	Product Availability & Revenue



OVERALL BUSINESS OUTCOME
Better Demand Planning + Stronger SKU Availability + Diversified Growth = Sustainable Revenue Scaling

Analysis Limitations & Strategic Next Steps

Current Analysis Limitations

Missing Customer Identifiers: A portion of transactions lack Customer IDs, limiting cohort retention analysis.

Absence of Margin Data: Dataset contains unit prices but no cost details, preventing net profitability metrics.

Bulk Order Skew: Extreme wholesale purchases introduce volatility into product averages.

Observational Limits: Historical transaction logs do not capture customer sentiment or promotional spend.

Recommended Next Steps

Integrate Cost & COGS Data: Model net contribution margin per product category.

Customer Segmentation: Conduct RFM analysis (Recency, Frequency, Monetary) on identified users.

Product-Level Seasonality: Map monthly demand curves for individual top revenue SKUs.

International Feasibility Study: Evaluate logistics and tax frameworks for top European targets.

Final Takeaway: Clear strategic opportunities exist in seasonal Q4 demand staging, SKU prioritization, and targeted European market diversification.